

Report on paper presentation on Superconducting diode effect in quantum spin Hall insulator based Josephson junctions[2]

Benedikt Scharf, Denis Kochan, Alex Matos-Abiague

Presented by Rishi Pareesh Joshi as a part of course on mesoscopic physics

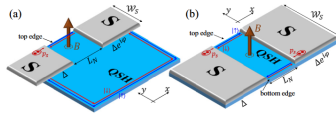


Figure 1. Topological Josephson junctions can be made using either (a) one or (b) two edges of a quantum spin Hall insulator (QSHI). In these setups, parts of the QSHI are covered by two s-wave superconductors that have different superconducting phases. Through the proximity effect, these regions have superconducting behaviour. The superconducting regions are denoted by the (S) parts, while the QSHI sections are called the normal (N) region. The normal region sits in a perpendicular magnetic field B . We set up our coordinate system so that the QSHI lies in the xy -plane, where x points along the direction current flows. The magnetic field points along the z -axis, or $\mathbf{B} = B\hat{e}_z$. We call the edge at positive y the top edge and the one at negative y the bottom edge. L_N is the distance between the superconducting regions, and P_S denotes the Doppler momentum gained by the superconducting quasi-particle condensate in the presence of a magnetic field[2].

Abstract—This report is a summary of the aim of the paper, explaining the work carried out in understanding the field and further, verifying some of the claims of the authors, and suggesting what the authors of the paper could have done more.

Keywords—*Superconductivity theory, Topological insulators, Superconducting diode effect theory, numerical codes and plots*

1. Aim of the paper Main

The aim of the paper was to scrutinise the magnetochiral properties of the two-dimensional (2D) Josephson junction (JJ) based on quantum spin Hall insulators (QSHI), as seen in Fig. 1. In particular, the paper focused on the Superconducting diode effect (SDE) when tuning an

- Out-of-plane magnetic field,
- Superconducting phase biasing,
- and Edge-channels asymmetry.

The effects of these parameters was explored by calculating the Josephson current and seeing the variation of the quality factor with magnetic field.

The paper also explores how, when the ground state parity is taken into consideration, the SDE effect is enhanced.

2. What did I do?

2.1. Covering the basics

I explored and covered the basics of superconductivity first from Prof. Saurabh Basu's NPTEL lectures. I studied from "Introduction to superconductivity" by M. Tinkham, the Bogoliubov de Gennes Hamiltonian, and further read about the Blonder-Tinkham-Klapwijk (BTK) formalism and scattering matrix approach to calculating Andreev Bound states (ABS) from research papers.

The basics of the quantum spin Hall effect were covered as part of classes, but additionally, I read about the topological aspects of these phenomena from the website topocondmat.org.

2.2. Understanding the paper

The model Hamiltonian was derived, giving importance to the key features of S-wave superconductors and the Cooper pairs in the presence of a magnetic field. The Hamiltonian is given for spin $s = \uparrow, \downarrow$ and edge $\sigma = t$ (top), b (bottom) as:

$$\hat{H}_{s,\sigma} = (s\sigma v_F \hat{p}_x - \mu_S) \tau_z + s \frac{v_F P_S}{2} + V_0 h(x) \tau_z + \Delta(x) [\tau_x \cos \phi_\sigma(x) - \tau_y \sin \phi_\sigma(x)]$$

Where:

1. v_F : Fermi velocity
2. $\hat{p}_x$: momentum operator
3. μ_S : chemical potential
4. $V_0 h(x)$: scalar potential profile
5. $p_S = \pi \hbar \frac{BW_S}{\phi_\sigma}$: Doppler shift momentum from orbital magnetic field B and width W_S
6. $\tau_{x,y,z}$: Pauli matrices in Nambu (particle-hole) space
7. $\phi_\sigma(x)$: superconducting phase at edge σ

I then derived the following things done in the paper:

1. Computed the ABS by using the ansatz wavefunction. I have also explained the choice for the ansatz wavefunction[1].
2. Calculated the continuum quasiparticle density of states using the scattering matrix approach in order to capture the contribution of the continuum density of states to the Free energy[1]. I found through numerical simulation that it was non-negligible.
3. Calculated the Free energy using the Krein-Friedel relation.
4. Found the expression for the Josephson current in the case where the parity constraint wasn't applied.
5. Simulated all the graphs for the parity unconstrained case and the parity constrained case (however, was not able to derive the parity constraint case from first principles).

3. What can be done more

- We can analyse the use of the SDE effect in a spin-selective rectifier setup and calculate its efficiency.
- We can understand the symmetries better, like time reversal symmetry, Inversion symmetry, and other spatial group symmetries, using a 3D topological insulator with S-N-S materials as layers.
- This model used a delta potential to approximate the QSHI, which has a length smaller than the coherence length for the superconducting material. We can use a Step-potential with the same length to model the QSHI-based S-N-S Josephson junction.

3.1. References

References

- [1] B. Scharf, A. Braggio, E. Strambini, F. Giazotto, and E. M. Hankiewicz, "Thermodynamics in topological josephson junctions", *Phys. Rev. Res.*, vol. 3, p. 033062, 3 Jul. 2021. DOI: [10.1103/PhysRevResearch.3.033062](https://doi.org/10.1103/PhysRevResearch.3.033062). [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevResearch.3.033062>.
- [2] B. Scharf, D. Kochan, and A. Matos-Abiague, "Superconducting diode effect in quantum spin hall insulator based josephson junctions", *Phys. Rev. B*, vol. 110, p. 134511, 13 Oct. 2024. DOI: [10.1103/PhysRevB.110.134511](https://doi.org/10.1103/PhysRevB.110.134511). [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevB.110.134511>.